



WallyGuard: Invention Analysis and Strategic Roadmap

This document serves as a comprehensive analysis and strategic roadmap for the WallyGuard device, a novel wearable communication and safety solution for non-verbal individuals and vulnerable adults. It details the core innovation, legal strategy, development plan, and multi-faceted value proposition.

1. Invention Disclosure: WallyGuard Device

The WallyGuard device is a closed-loop wearable communication and safety device intended to prevent abuse and neglect by providing objective documentation of distress or adverse events for non-verbal individuals (e.g., severe autism) and vulnerable adults (e.g., dementia, nursing home residents).

A. Problem and Core Innovation

Category	Description
Problem to Solve	Abuse and neglect of non-verbal individuals in care/school settings due to lack of an objective witness and accountability. Existing surveillance devices (e.g., Angel Sense) are banned in schools due to privacy laws (e.g., COPPA, Wiretapping).
Core Solution	A "closed-loop" wearable system that detects distress patterns (visual or biometric) and sends a discreet, timestamped alert/metadata

Category	Description
	to the caregiver/guardian, without recording or storing raw audio or visual data.
Legal Strategy (Value)	Classify the device as a "communication device" protected under the Americans with Disabilities Act (ADA) , forcing schools to allow it as a reasonable accommodation for FAPE.
Current Development Stage	Concept and Legal Strategy Development. Ready for provisional patent filing and pilot program.

B. Suggested Initial Next Steps

The immediate actions required to secure the idea and move toward a prototype are:

- 1. **Protect the Idea:** File a **provisional patent application** (DIY if needed, \$75) and finalize detailed concept documentation (date, sign, and witness concept).
- 2. **Seek Legal Coverage: Contact Protection and Advocacy (P&A)** for free legal guidance on the ADA vs. Surveillance classification.
- 3. **Validate Concept:** Begin a **pilot with Walter** to gather real data and proof of concept.

2. Product Analysis and Feature Roadmap

The WallyGuard device is structured around a "closed-loop" design to mitigate legal risks while maximizing utility for its target populations.

A. Core "Closed-Loop" Technical Design

This design is the core legal workaround to classify the device as communication, not surveillance.

Feature	Technical Specification	Legal Justification
Audio Processing	CLOSED LOOP: Audio is processed locally on the device (e.g., detecting a "help,	Avoids state Wiretapping Laws and reduces COPPA risk by

Feature	Technical Specification	Legal Justification
	help now" phrase or a prolonged distress sound pattern).	preventing the recording of raw, two-party conversation.
Data Recording	NO RAW AUDIO/VIDEO STORED: Explicitly designed to NOT record or store raw audio or visual footage.	Protects against classification as a "recording/surveillance" device and mitigates FERPA risk for other students' privacy.
Transmission	ALERT-ONLY METADATA: Only transmits metadata (time, location, trigger type, biometrics) to the caregiver/legal guardian.	Establishes the function as " communication of distress " rather than surveillance data collection.

B. New Design and Functionality Considerations

Functional Area	Feature Description	Target Population Value
Detection Refinement	Include supplementary biometric data like pulse and heart rate to confirm distress signals detected by visual patterns.	Provides objective, undeniable proof of a distress event for legal accountability.
First Responder Alert	Simple chip/signal integrated into the wearable that first responders (police, EMTs) can scan.	STOP, THINK, OBSERVE alert to prompt immediate de-escalation for a vulnerable person.
Form Factors	Offer multiple sensory-friendly designs: Watches, small pins, luggage-type tags, AirTag-like tags, and ankle placement.	Ensures maximum user compliance and accommodation of sensory issues in non-verbal children.
Market Pivot	Visual Detection Only (No Audio): This pivot is required if wiretapping laws kill the audio feature.	Mitigates the highest legal risk while still providing location and distress monitoring.

3. Value Proposition and Market Analysis

The WallyGuard device offers substantial value across legal, ethical, and commercial domains, targeting three primary populations.

A. Target Populations

The device targets the following vulnerable groups:

- **Non-verbal individuals** (e.g., severe autism, TBI) in schools and care facilities.
- **The elderly with Alzheimer's/dementia** and other cognitive impairments.
- **Nursing home residents** and vulnerable adults in high-risk care facilities.

B. Value Assessment Matrix

The WallyGuard device provides value in all major categories, especially by leveraging a strong legal shield (ADA).

Value Category	Assessment	Score (1-5)	Weight	Total Score
Patentability (Novelty/Non-Obviousness)	High (The "closed-loop" legal workaround is highly novel)	5	20%	1.00
Technical Feasibility	Moderate (Biometrics/visual detection is proven, integration is the challenge)	4	15%	0.60
Commercial Market Size	High (Global market: Autism Tech, Elder Care, Nursing Home Safety)	5	25%	1.25
Legal/Ethical Value (ADA/FAPE)	Critical (Addresses systemic failure; provides accountability and FAPE compliance)	5	30%	1.50
Competition Advantage	High (Superior to Angel Sense by mitigating school ban risk)	4	10%	0.40
Total Weighted Value Score			100%	4.75 / 5.00

C. Funding and Manufacturing Plan

Initial funding will be secured through grants and existing revenue streams, followed by a targeted partnership strategy.

Phase	Strategy	Key Resources/Partners
Phase 1: Seed Funding	Use grants and existing revenue for R&D and pilot programs.	Grants: NIH SBIR/STTR, Autism Speaks tech grants, NSF assistive technology, State disability innovation funds. Personal: Content Creation (Wally) revenue.
Phase 2: Legal Shield	Secure legal classification as a "communication device."	Protection and Advocacy (P&A) agency.
Phase 3: Manufacturing	Find a partner with existing regulatory or manufacturing capacity.	Engineering Partner (to build prototype) or Existing DME Manufacturer (if FDA classification is unavoidable).
Phase 4: Scaling	Target large-scale partnerships for distribution in health and education.	Autism Speaks , & Person (for corporate partnership, e.g., Google's tech/brand), State Departments of Health/Education .

4. Legal Conflict Analysis: ADA vs. Surveillance

The success of the WallyGuard device hinges entirely on winning the legal battle over its classification.

The core legal conflict is defined by the following question: **Is a wearable device that detects distress and alerts caregivers/legal guardians a "communication device" protected under the Americans with Disabilities Act (ADA), or a "recording/surveillance device" restricted under COPPA/privacy laws?**

Side of the Conflict	Legal Argument	Rationale and Law
WallyGuard Position (ADA/Communication)	The device is a necessary Augmentative and Alternative Communication (AAC) aid that provides a means of communication (distress) for a non-verbal user.	Americans with Disabilities Act (ADA) and the right to Free Appropriate Public Education (FAPE) mandate that schools must provide reasonable accommodation, which includes communication devices.
Opposing Position (Surveillance/Recording)	The "always listening" (audio or visual processing) function, and the transmission of metadata, legally constitutes surveillance and data collection.	Children's Online Privacy Protection Act (COPPA) , State Wiretapping Laws (Two-Party Consent), and FERPA (Privacy of other students).
Strategic Solution	The " closed-loop " technical design is the legal workaround, arguing that because no raw audio/video leaves the device , it is not a recording device, but an alarm or communication trigger.	The device's function is to ensure accountability and provide an objective witness when staff/paraprofessionals are unable or unwilling to prevent harm.